The capital A is important here to distinguish the principal value from the general value.

Evidently, in practice to find the principal angle θ, we usually compute $\alpha = \tan^{-1}\left|\frac{y}{x}\right|$ and adjust for the quadrant problem by adding or subtracting α with π appropriately.

$$\arg z = Arg\ z + 2n\pi,\ n \in \mathbb{Z}.$$

Some of the properties of arguments are

(1) $\arg(z_1\ z_2) = \arg z_1 + \arg z_2$

(2) $\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$

(3) $\arg(z^n) = n\ \arg\ z$

(4) The alternate forms of $\cos\theta + i\sin\theta$ are $\cos(2k\pi+\theta) + i\sin(2k\pi+\theta)$, $k \in \mathbb{Z}$.

For instance the principal argument and argument of $1, i, -1$, and $-i$ are shown below:-

z	1	i	-1	$-i$
$Arg(z)$	0	$\frac{\pi}{2}$	π	$-\frac{\pi}{2}$
$\arg z$	$2n\pi$	$2n\pi + \frac{\pi}{2}$	$2n\pi + \pi$	$2n\pi - \frac{\pi}{2}$

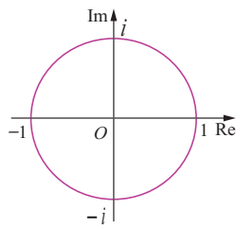


Fig. 2.34

Illustration

Plot the following complex numbers in complex plane

(i) $5\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$

(ii) $4\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$

(iii) $3\left(\cos\frac{-5\pi}{6} + i\sin\frac{-5\pi}{6}\right)$

(iv) $2\left(\cos\frac{\pi}{6} - i\sin\frac{\pi}{6}\right)$.

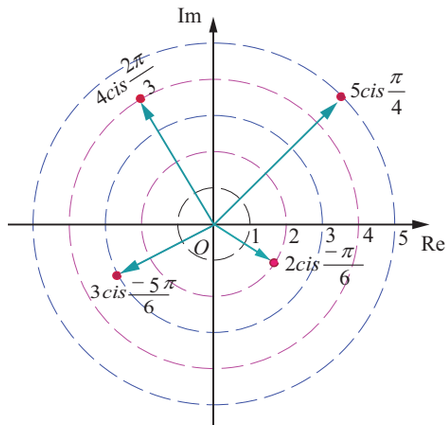


Fig. 2.35

2.7.2 Euler's Form of the complex number

The following identity is known as Euler's formula

$$e^{i\theta} = \cos\theta + i\sin\theta$$

Euler formula gives the polar form $z = r\,e^{i\theta}$

Note

When performing multiplication or finding powers or roots of complex numbers, Euler form can also be used.

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